

# Water Softener & Ion-Exchange Systems

*High-Capacity Hardness Removal and Scaling Prevention.*

Softeners and ion-exchange systems

## Technical Specifications

| Parameter                                                                          | Specification / Value |
|------------------------------------------------------------------------------------|-----------------------|
| Typical preliminary value                                                          | Resin type            |
| Strong-acid cation resin                                                           | Inlet hardness        |
| Commonly up to approximately 1,000 mg/L as CaCO <sub>3</sub> , subject to supplier | Outlet hardness       |
| Typically <5 mg/L as CaCO <sub>3</sub> ; lower values may require polishing        | Service flow          |
| Approximately 10-40 BV/h                                                           | Resin bed depth       |
| Approximately 0.75-1.5 m                                                           | Backwash              |
| 10-20 min                                                                          | Brine draw            |
| 20-60 min                                                                          | Slow rinse            |
| 20-60 min                                                                          | Fast rinse            |
| 10-30 min                                                                          | Regenerant            |
| Sodium chloride                                                                    | Regeneration trigger  |
| Volume treated, hardness leakage or time                                           | Required pretreatment |
| Turbidity, iron, manganese and oil removal                                         |                       |